

Komodo Dragon Element B

While many previous attempts are not on the internet, one attempt at specifically catching Komodo Dragons was documented in the Reptiles Magazine. Experts from places like the Komodo National Park and the Woodland Park Zoo came together in Indonesia to catch and release the dragons. The main zookeeper was a herpetologist from the Turtle Back Zoo in West Orange, N.J., who previously had worked with komodo dragons at the Memphis Zoo. The dragon that this zookeeper was catching was around 8 feet and 150 pounds, much smaller than the male in the Nashville Zoo, at 10.2 feet long and 231 pounds. The American Zoological Association donates money to the conservation project for the dragons. In return, 2 of their workers are sent to Indonesia to participate in the catch-and-release process.



The traps that these zookeepers used to catch the dragons were three boxes, each 3 feet long, with rotten goat meat on the inside hanging by a wire. When the dragon entered the box

and pulled on the meat, also pulling the wire, a pin would be yanked free. The pin holds the sliding door open.

According to the zookeeper, releasing the dragons from the trap was much more difficult than getting them to enter. The traps had mini openings on the top of each 3-foot section, which would slide open for a rope to be lowered and wrapped around the dragon. The release is described as the head, by the noose, being pulled to the top of the cage while the tail and back legs were pulled out and the mouth taped shut.



The author of the article states that he and his group don't use anesthesia to catch the dragons because of the cost but also the risk of causing harm to the dragon. This is similar to how our box needs to be because we cannot do anything to the dragon; we simply just trap it in the box. Once the dragons are given the anesthesia, it can take several hours to work. One of the things that makes this dangerous for the dragons is that they can remain unconscious for multiple

days. The author claims that sometimes, when they are unconscious, they need someone to manually help them breathe.

In another one of their attempts to capture a dragon, they ended up not using the trap and doing it with just a rope. They lowered the noose to the dragon's head and while one person grabbed the tail, another tied its legs.

Many aspects of their design proved successful, and many of them are similar to the requirements needed for our zoo project. One part is the bait they used. They explained how the Komodo dragons are very attracted to the smell of rotten goat meat. Another successful part was the idea to attach the bait to a wire at the very end of the trap that could close the door. This makes sure that the dragon is the entire way into the cage and won't be in the way when the door closes. The release from the cage was unsuccessful for this attempt. Needing multiple people, 3 in their case, to take the dragon from the cage and also having to disassemble the trap when the dragon was too large were the difficult aspects of this cage. Their cage did not allow testing and only had mini doors for the nooses. Information about the handles and the overall design of the frame was not included.

Link: Komodo Dragon [previous attempt](#)

The information for previous attempts is incredibly scarce as there seem to be only images. However, there are alternatives, specifically an alligator cage. All images documenting the Komodo dragon cages share three main things in common. The first is that it is normally a wooden box with only one entrance. The next element they share is a see-through roof. The third thing in common is that the box only allows forward and backward movements, which restricts the Komodo dragon from moving inside the cage. While there is only visual information for the Komodo dragon cages, there is a lot more on alligator cages. Alligator cages are set up the same

way the cages are set up, being long rectangular boxes that prevent movement to the side.

However, unlike the Komodo Dragon cages, they are fences made out of metal and have a longer gate system at the front of the cage, incredibly similar to a guillotine. Alligator cages are typically transported with backpack straps, while the Komodo Dragon cage is normally carried with basic handles in the same vein as a coffin.

One attempt at a box similar to a Komodo dragon is an alligator trap. The alligator trap is inspired by bear and deer traps. Officers who work with the N.C. Wildlife Resource Commission built a trap that was 10 feet long and 24 inches in diameter. The ends had steel grates that closed on both sides. To use this, the officers place a rope around the alligator and pull the rope through the tube. This allows the alligator not to be touched but to be transported. The article states that the tube also eliminates the ability of the alligator to turn around.



The success that came with this design is that the size of the box does not matter because the size cannot change the ability to turn around. Also, this tube allows for the animal to be transported easily. However, the tube does not allow for testing and only transportation. Along with that, the risk of injury is still high because of the need to place a rope around the alligator.

Link: [alligator trap past attempt](#)

One previous attempt for handles is the Dynasty Hardware European Bar. This is a simple cabinet handle that is 5-3/4".

Link: [dynasty handle](#)



pros	cons
comfortable	Only for light weight
simple	
Appealing to the eye	

cheap	
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Analysis:

This handle is small, but at a larger scale, the product would fit nicely into the users' hands.

However, not being made with a durable material leaves the uncertainty that it will break when lifting heavy. This handle is suitable for small projects, but it will not be beneficial for larger, heavier projects. It successfully appeals to the eyes and is cheap and easy to use.

Another attempt at handles is the Home Depot Amerock Cup Pulls. These handles are 3-inch, or 76mm.

Link: [amerock handle](#)



pros	cons
Smooth, sleek, appealing look	Only for light weight
simple	semi-comfortable
cheap	

Analysis: A small handle like this one is beneficial for small, lightweight products to lift, like cabinets. This would be unsuccessful for our project because it is not made with the strong metal that is needed for the weight of the box and dragon. The semi-circular shape with a closed top only allows for hands of certain sizes to be able to use the handle. Lots of weight also would hurt the user's hands because of a surplus of weight on one section of the hand.

A more heavy-duty handle more suitable for a cage is one like the stainless steel industrial one.

This specific handle is from Amazon, and the measurements are 46mm tall and 200mm long.

Link: [handle](#)

pros	cons
Appealing to the eye	expensive

Simple	
durable	
comfortable	

Analysis: The material used, stainless steel, allows for a much larger amount of weight to be applied. While some handles are movable, light, or complex, a simple, durable handle will be the best to apply to our alligator cage. This is because it won't break as the animal is being transported, and it is easy to install and use. However, even though it is expensive, the safety of the animal and the trainers, and their hands, is more important.